

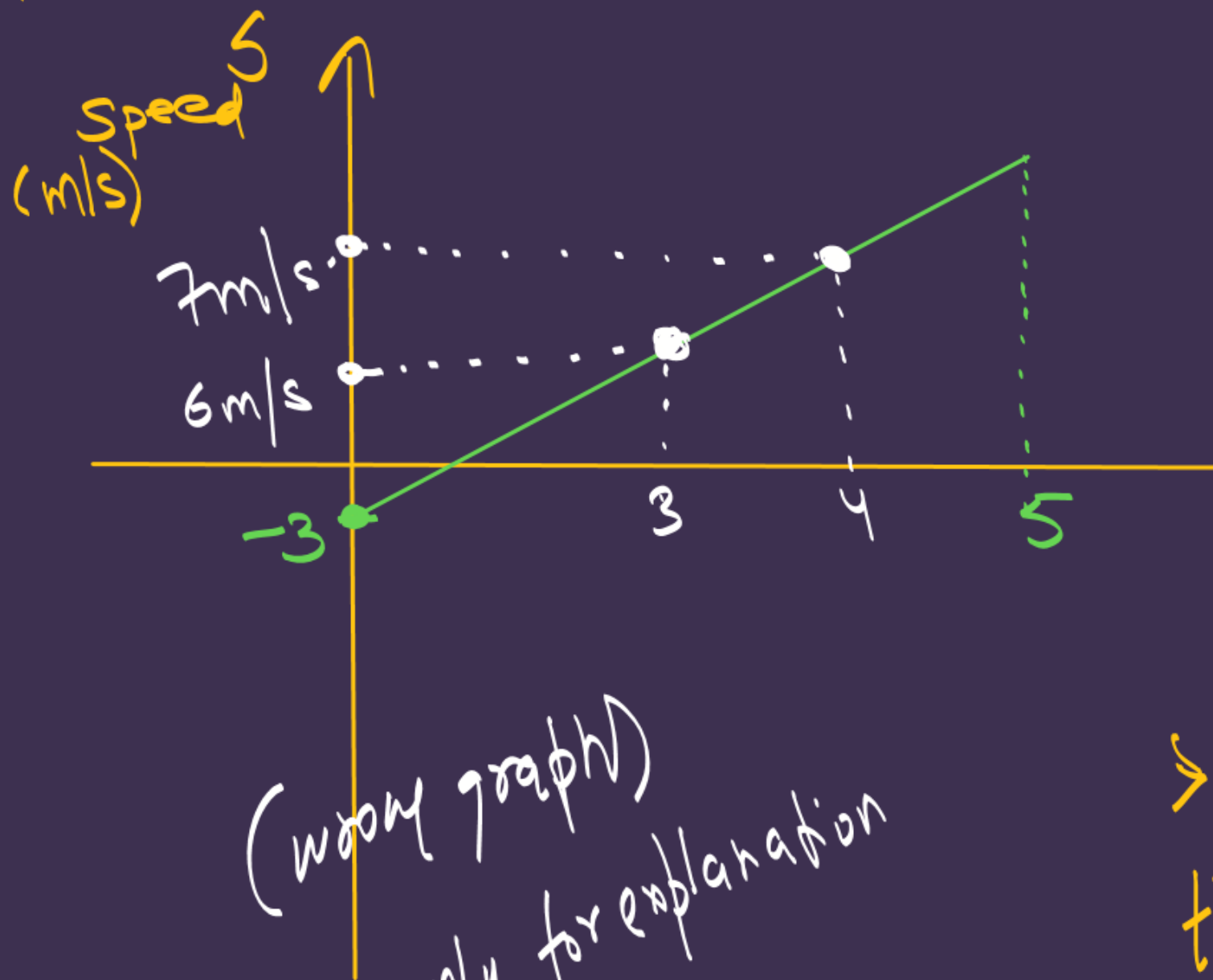
[Aim: 100/100 in Maths]

आरंभ

POLYNOMIALS

Lecture #7

#(K3D) :- GRAPH की जानकारी :-



Q A car is moving by the equation
 $S = 3 + t$

(i) What is the speed of car at 4th second?

$$t = 4 \rightarrow S = 3 + 4$$
$$S = 7 \text{ m/s}$$

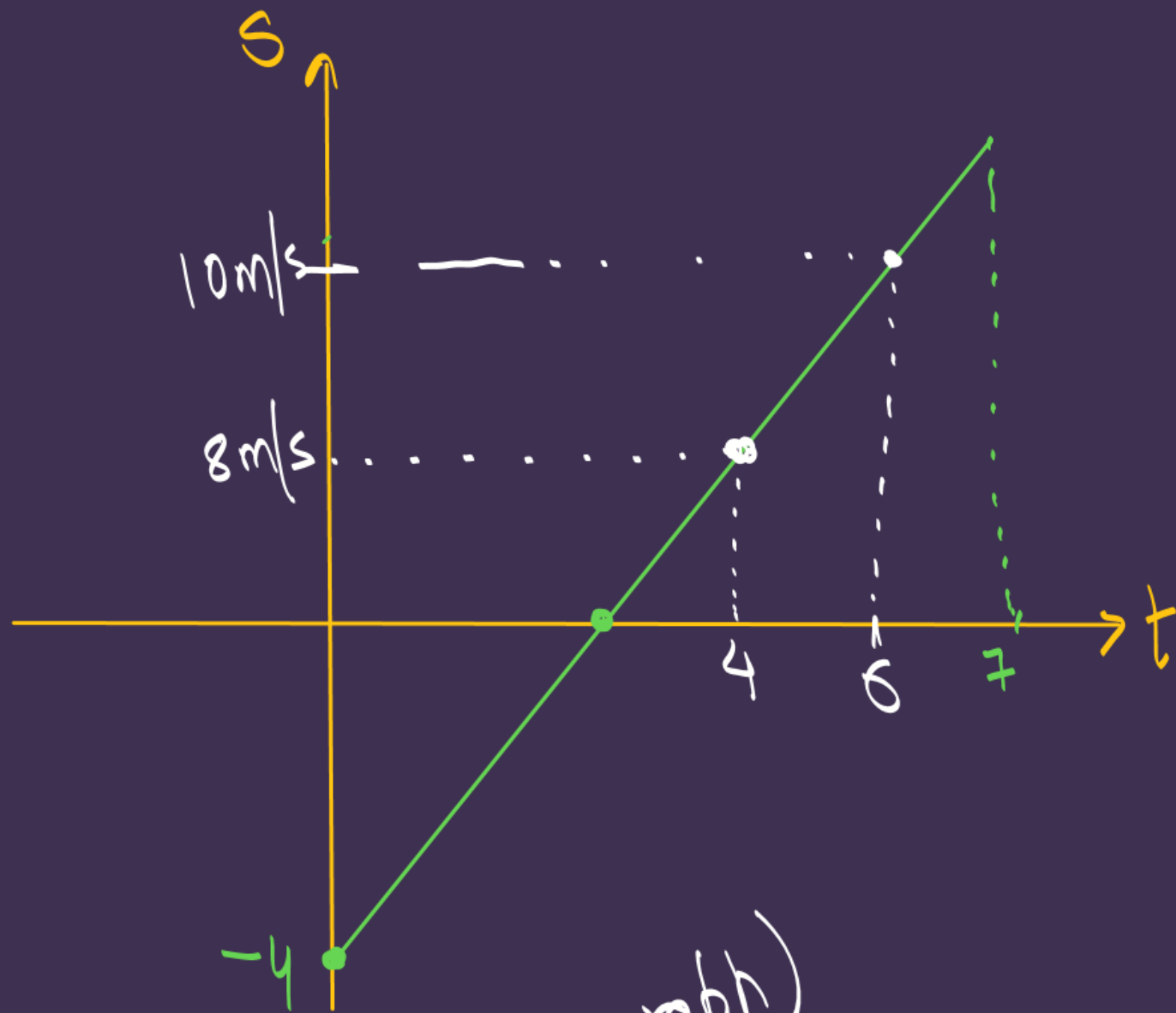
(ii) at $t = 3 \text{ sec}$, $S = ??$

$\rightarrow t$
time (sec)

$$S = 3 + 3$$
$$S = 6 \text{ m/s}$$

(iii) at what time speed of car is 2 m/s

$$2 = 3 + t \Rightarrow t = 2 - 3 = -1 \text{ sec}$$



(wrong graph)

Speed of car is given by

$$S = 4 + t$$

(i) Find speed at 6th second?

$$t = 6 \Rightarrow S = 4 + 6$$

$$S = 10 \text{ m/s}$$

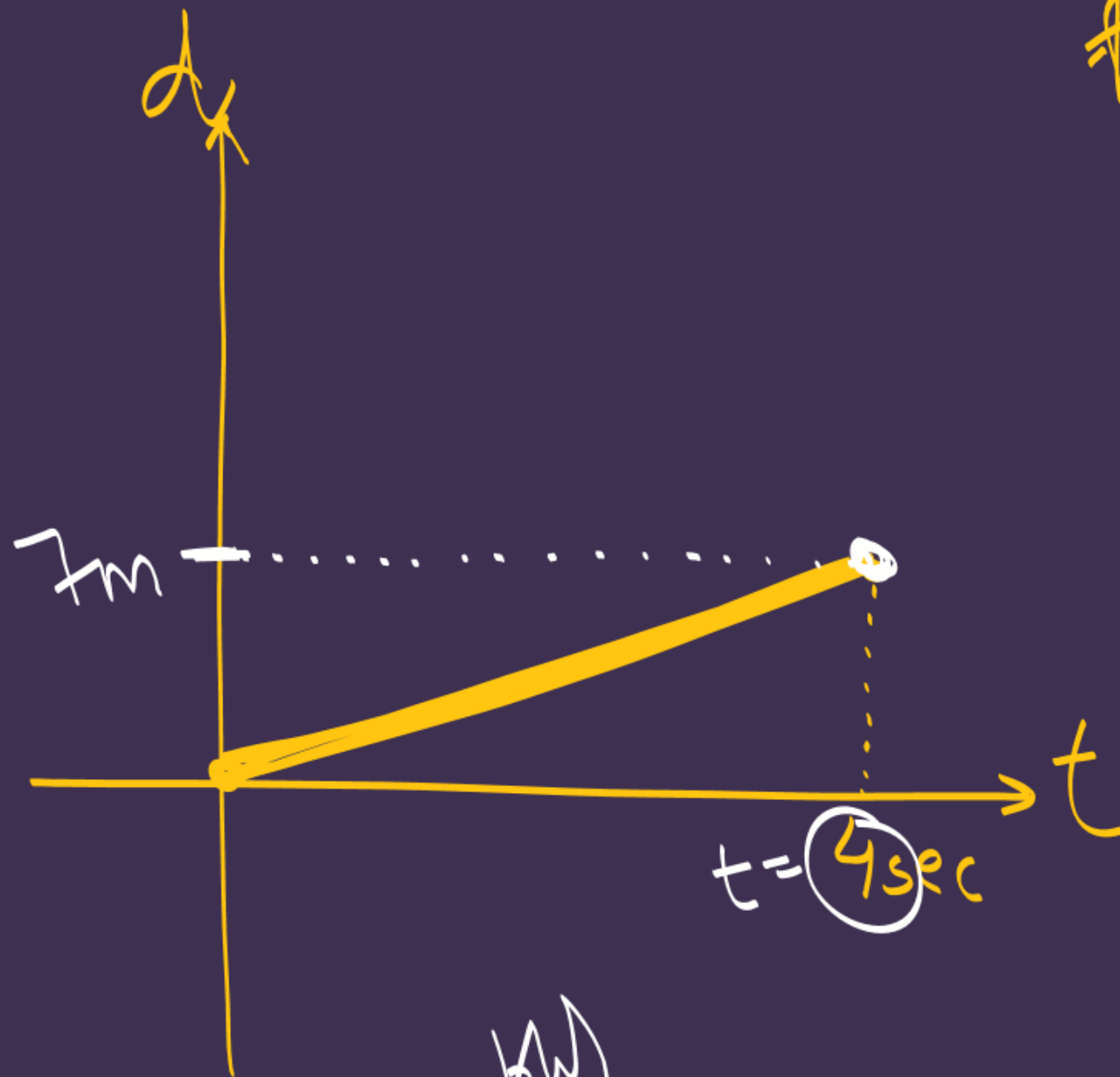
(ii) Find time at which speed of car is 8 m/s.

$$S = \underline{8 \text{ m/s}}$$

$$8 = 4 + t$$

$$8 - 4 = t$$

$$t = 4 \text{ sec}$$



(wrong graph)

For a car distance is given by
 $d = 3 + t$

Q find the max distance travelled by the car
we can clearly see at $t = 4\text{sec}$, distance was max.

$$d = 3 + (4)$$

$$d = 7\text{m}$$

Q A car is moving by the eq $\Rightarrow y = t(4-t)$ y is the distance covered by the car.

(i) At what time car was at max distance?

$t = ?$

$d = d_{\max} = 4m$

put

$$4 = t(4-t)$$

$$t = 2 \text{ sec}$$

(ii) What is the distance covered by car at 3rd second?

$$d.?, t=3 \quad d = 3(4-3)$$

$$d = 3m$$

(iii) At what time distance was 3m?

$$d = 3m$$

$$t = ?$$

$$3 = t(4-t)$$

$$3 = 4t - t^2$$

$$t^2 - 4t + 3 = 0$$

$$t^2 - 3t - t + 3 = 0$$

$$t(t-3) - 1(t-3) = 0$$

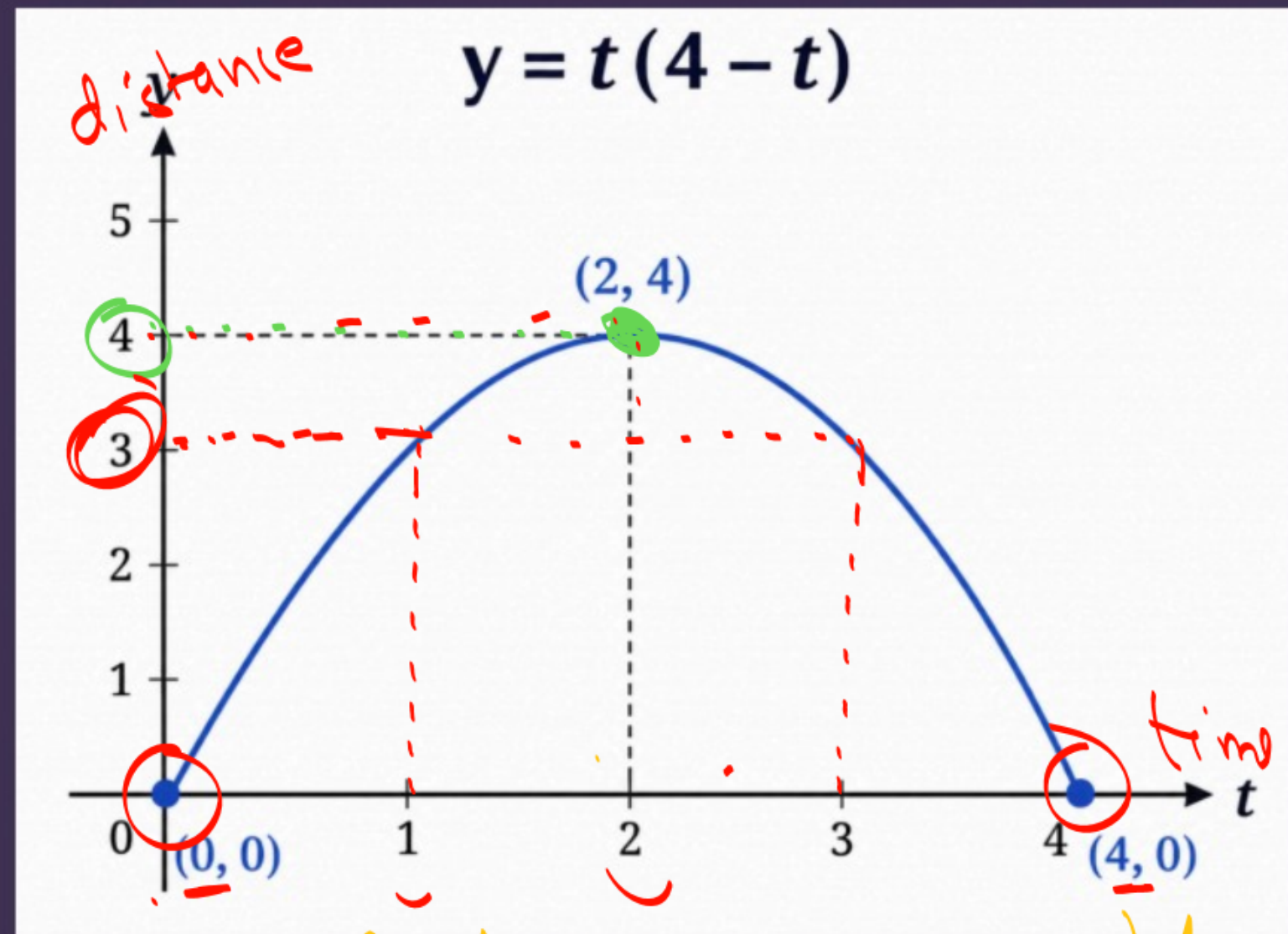
$$(t-3)(t-1) = 0$$

$$t = 3 \text{ sec}$$

$$t = 1 \text{ sec}$$

(iv) Find the zeroes of this polynomial!

$$\{0, 4\}$$



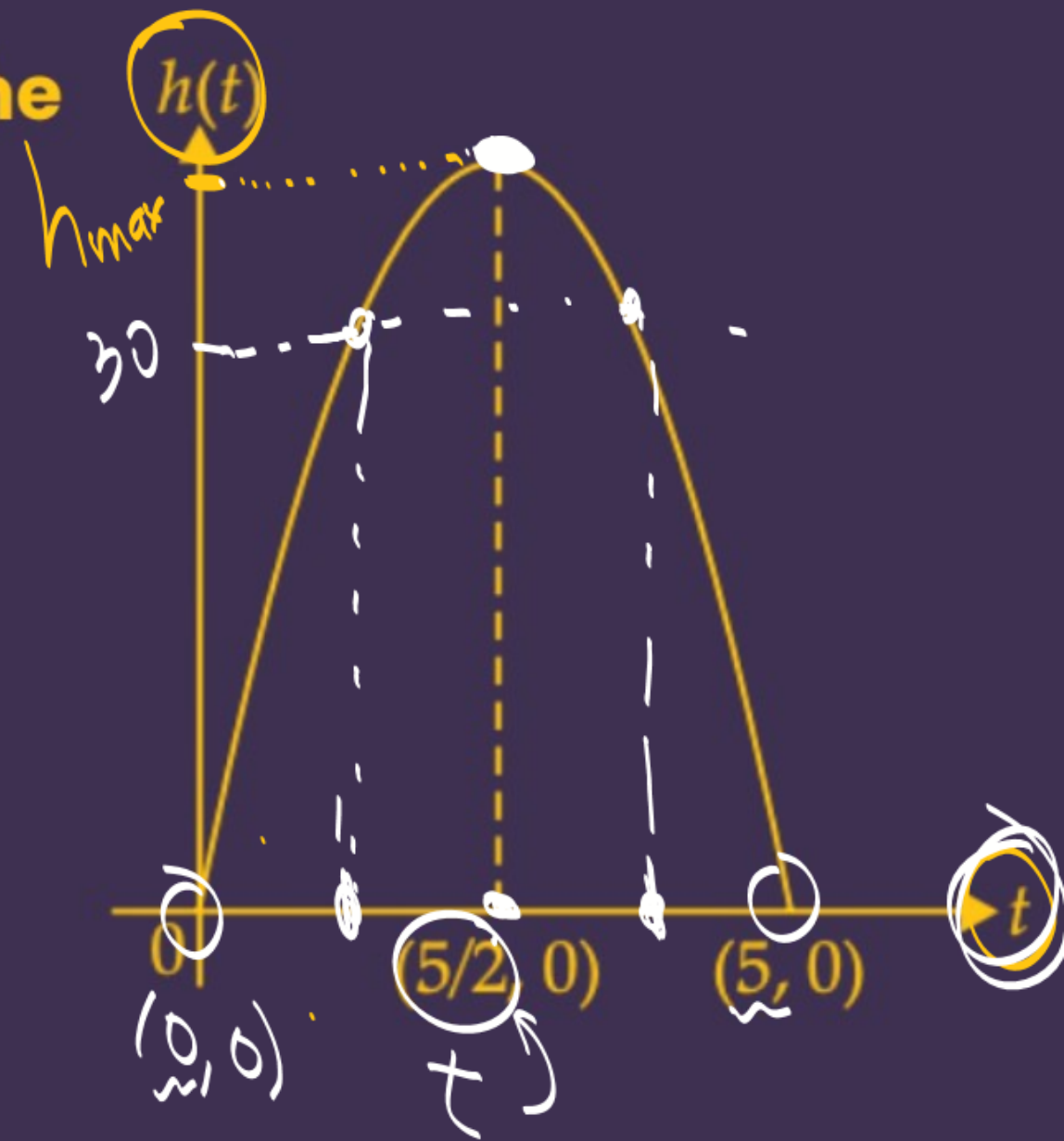
#LP: A ball is thrown in the air so that t second after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$.

Observe the graph of the polynomial and answer the following questions:

(i) Write zeroes of the given polynomial. [1]

$$h = 25t - 5t^2$$

0 & 5



(ii) Find the maximum height achieved by the ball.

$$h = ?$$
$$t = \frac{5}{2}$$

put $\rightarrow$

$$h = 25\left(\frac{5}{2}\right) - 5\left(\frac{5}{2}\right)^2$$
$$= \left(\frac{5}{2}\right) \left[25 - 5 \times \frac{5}{2} \right]$$

$$h = 25t - 5t^2$$

$$\rightarrow \frac{5}{2} \left(25 - \frac{25}{2} \right)$$
$$\rightarrow \frac{5}{2} \left(\frac{50 - 25}{2} \right)$$
$$\rightarrow \frac{5}{2} \times \frac{25}{2} = \frac{125}{4} \text{ m}$$

$\rightarrow 31.25$

(iii) (a) After throwing upward, how much time did the ball take to reach to the height of 30 m?

OR

(b) Find the two different values of t when the height of the ball was 20 m

$$t = ?$$
$$h = 30 \text{ m}$$

$$\text{put } \rightarrow 30 = 25t - 5t^2$$
$$5t^2 - 25t + 30 = 0$$
$$5(t^2 - 5t + 6) = 0$$

$$t^2 - 5t + 6 = 0$$
$$\rightarrow t^2 - 3t - 2t + 6 = 0$$
$$\rightarrow t(t-3) - 2(t-3) = 0$$
$$\rightarrow (t-3)(t-2) = 0$$

$$t - 3 = 0$$
$$t = 3 \text{ sec}$$
$$t - 2 = 0$$
$$t = 2 \text{ sec}$$

The shape of the curve shown below can be represented by the polynomial $p(x) = -x^2 + 2x + 8$, where x is the length (in feet) on stage level. Based on the figure given above, answer the following questions:

i. Determine the height of the arch.

$$y = -x^2 + 2x + 8$$

height $\rightarrow y$ की length

put $x = 1 \text{ m}$

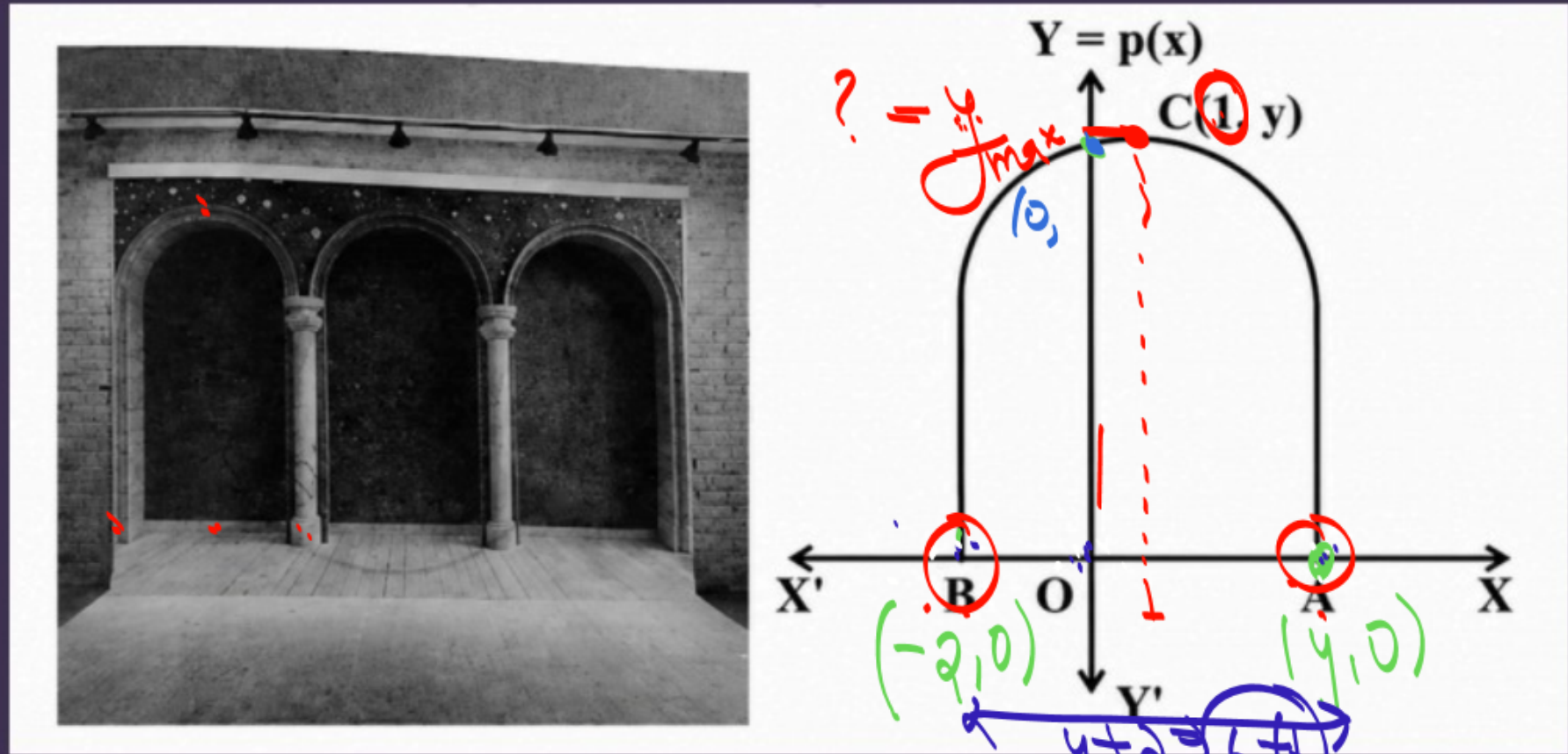
$$y = -(1)^2 + 2(1) + 8$$

$$y = -1 + 2 + 8$$

$$y = t + 1 + e$$

$$y = 9$$

9ft height



ii. a. Find zeroes of the polynomial $p(x)$. Which points on the graph represent the zeroes?

width OR

b. Find the span of the arch on the stage floor. $\rightarrow 6 \text{ ft}$

$$-x^2 + 2x + 8 = 0$$

$$-x^2 + 4x - 2x + 8 = 0$$

$$-x(x-4) - 2(x-4) = 0$$

$$(x-4)(-x-2) = 0$$

$$x-4=0$$

$$x=4$$

$$-x-2=0$$

$$x=-2$$

iii. Write the coordinates of the point of intersection of the above curve with the y-axis.

at all point of y-axis $\Rightarrow x=0$

$$\therefore \text{put } x=0 \Rightarrow y = 0 + 0 + 8$$

$$y=8$$

$$\therefore \text{coordinates } (0, 8) \checkmark$$

#LP: An arch of a railway bridge, built on Chenab riverbed, is shown in the above diagram. It is a parabolic arch connecting two hills at P and Q.

If the parabola curve is represented by the polynomial

$$p(x) = -0.0025x^2 - 0.025x + 136$$

COSE 2026

Observe the diagram and based on above information, answer the following questions:

(i) Write the co-ordinates of point A.

$$y = -0.0025x^2 - 0.025x + 136$$

(i)

A → y axis

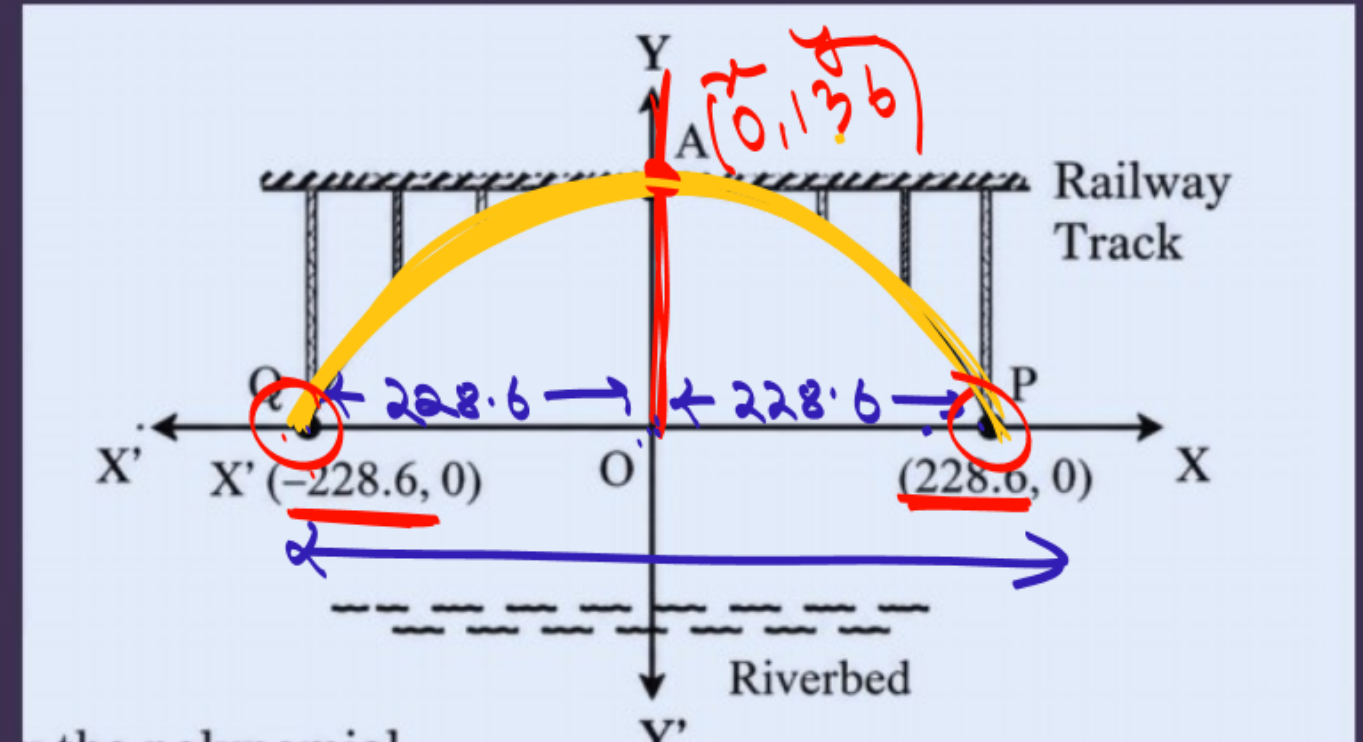
we know $x=0$

put $x=0$

$$y = 0 + 0 + 136$$

$$y = 136$$

$$A(0, 136)$$



(ii) Find the span of the arch.

width

$$228.6 + 228.6$$

$$457.2$$

$$-228.6 \text{ and } +228.6$$

(iii) (a) Write the zeroes of the polynomial using diagram and verify the relationship between sum of zeroes and polynomials.

OR

(b) Find the values of $p(x)$ at $x = 100$ and $x = -100$. Are they same?

put

put

जो variable find करवा है
उसके दूसरे वाला dhond lo!

Polynomials अपराध!!

THANK YOU COODIESS !!

HW → Complete Module

